LexAI: Intelligent Legal Assistance System

Project Charter · LEXAI-2026-SPRING · March 29, 2026

1. Project Overview

Item	Detail
Project Title	LexAI: Intelligent Legal Assistance System
Project ID	LEXAI-2026-SPRING
Project Type	Course Project Prototype
Start Date	March 30, 2026
Finish Date	June 14, 2026
Project Manager	Xiangyu Xiao
Charter Date	March 29, 2026
Faculty Advisor	Bowen Du

2. Stakeholder Analysis

Name	Role	Contact	Focus
Xiangyu Xiao	Team Leader / PM	2351110@tongji.edu.cn	Architecture, AI integration, full-stack dev, milestone control
Baoyi Hu	Team Member	2353409@tongji.edu.cn	Model-service refinement, output-logic tuning
Sirui Da	Team Member	2352288@tongji.edu.cn	Workflow organization, interaction polishing
Qi Lin	Team Member	2352609@tongji.edu.cn	Interface integration, testing, deployment, delivery
Bowen Du	Faculty Advisor	—	Academic guidance, milestone review
TBD	Legal Reviewer	—	Legal-logic review for templates and disclaimers
Ext. Pilot Users	Pilot Users	—	Pilot testing and usability feedback

3. Project Description

3.1 Background

LexAI explores AI-assisted legal-service workflows in a controlled prototype setting. It targets the uneven distribution of professional legal support in China and leverages large language models and RAG-based retrieval. The prototype is scoped to Chinese civil law (PRC Civil Code) for academic demonstration only.

3.2 Problem Statement and Opportunity

Item	Description
Challenge 1	Professional legal services are unevenly distributed across regions and user groups.
Challenge 2	Individuals and small organizations face high consultation and document-preparation costs.
Challenge 3	Repetitive retrieval, drafting, and review tasks are well-suited for AI-assisted support.
Opportunity	A prototype can verify AI-assisted legal workflow feasibility before broader deployment.

3.3 Target Outcomes

Impact Area	Desired Impact
Efficiency	Reduce time for representative legal consultation and drafting tasks.
Quality	Improve consistency and traceability of prototype-generated outputs.
Demonstration	Deliver a campus-scale prototype: consultation, case analysis, contract review, doc generation.
Foundation	Build a reusable technical and evaluation base for later iterations.

4. Measurable Organizational Value (MOV)

Objective	Metric	Baseline	Target (Jun 2026)
Task efficiency	Avg. completion time (representative tasks)	100% manual	≥ 20% reduction
Output quality	Review checklist pass rate	75%	≥ 85%
Contract risk detection	Recall on benchmark set	65%	≥ 80%
Accessibility	Pilot users completing ≥ 1 workflow	0	≥ 20 users
Workflow stability	Core workflow success rate	85%	≥ 95%
Hallucination control	Unsupported claim rate (audited samples)	15%	≤ 8%

5. Project Scope

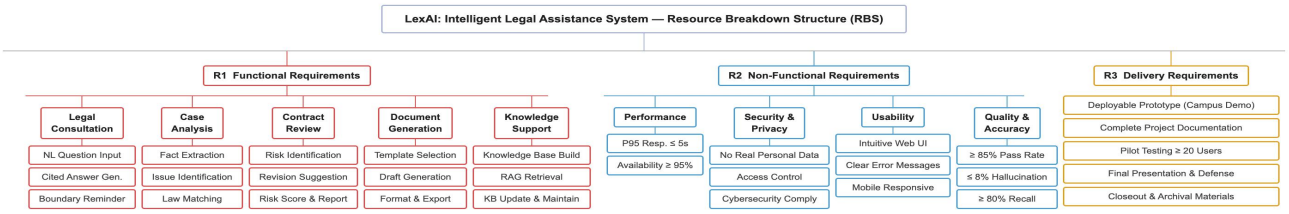
5.1 In-Scope Items

Item	Description
Legal consultation	NL question input, cited structured answers, boundary reminders
Case analysis	Fact extraction, issue identification, applicable-law matching
Contract review	Clause-level risk identification, revision suggestions, risk scoring
Document generation	Template selection, draft generation, format validation and export
Knowledge support	Legal KB construction, RAG-based retrieval, KB maintenance
System delivery	Demo deployment, pilot testing, logging, evaluation records

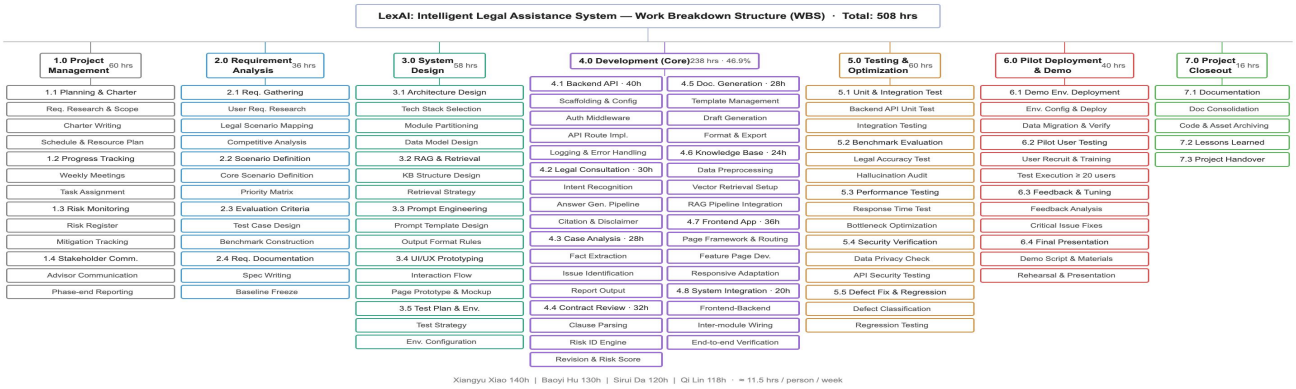
5.2 Out-of-Scope Items

Item	Description
Formal legal representation	No court, arbitration, or dispute-handling
Binding legal advice	Outputs require professional review before use
Commercial operation	No production-grade or 24/7 public deployment
Full legal coverage	Scoped to Chinese civil law only

5.3 Requirement Breakdown Structure (RBS)



5.4 Work Breakdown Structure (WBS)



5.5 Work Hours Summary

Phase	Hrs	%	Xiangyu	Baoyi	Sirui	Qi Lin
1.0 Project Management	60	11.8%	30	10	10	10
2.0 Requirement Analysis	36	7.1%	8	10	10	8
3.0 System Design	58	11.4%	16	22	12	8
4.0 Development	238	46.9%	50	64	64	60
5.0 Testing	60	11.8%	16	14	14	16
6.0 Pilot & Demo	40	7.9%	12	6	10	12
7.0 Closeout	16	3.1%	8	4	0	4
Total	**508**	**100%**	**140**	**130**	**120**	**118**

~127 hrs/person over 11 weeks ≈ 11.5 hrs/person/week.

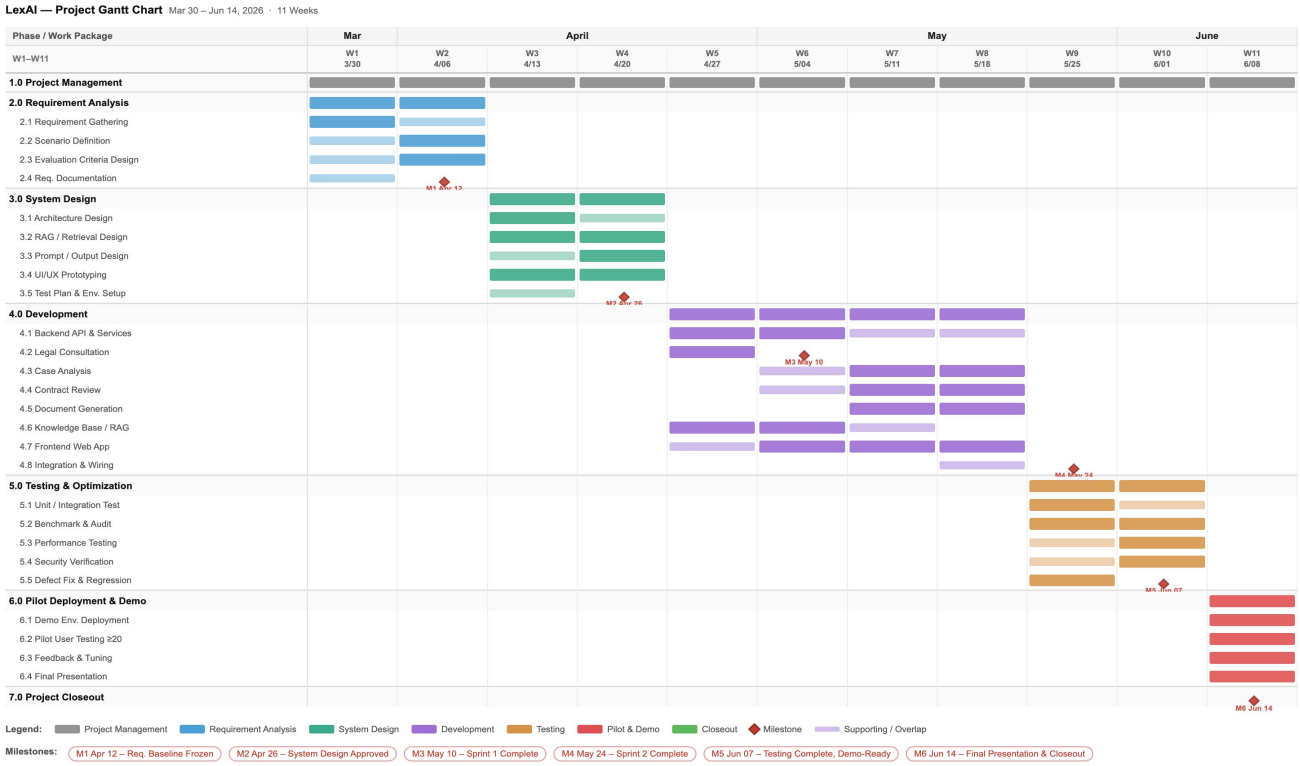
6. Project Schedule

6.1 Phase and Milestone Plan

Phase	Weeks	Dates	Key Deliverables	Milestone
Requirement Analysis & Planning	W1–W2	Mar 30 – Apr 12	Specs, scenario list, evaluation criteria, charter	M1: Req. baseline frozen
System Design	W3–W4	Apr 13 – Apr 26	Architecture, RAG strategy, prompt design, UI prototype	M2: Design approved
Development – Sprint 1	W5–W6	Apr 27 – May 10	Backend, knowledge base, consultation module	M3: Sprint 1 complete
Development – Sprint 2	W7–W8	May 11 – May 24	Case analysis, contract review, doc gen, frontend	M4: Sprint 2 complete

Phase	Weeks	Dates	Key Deliverables	Milestone
Integration & Testing	W9–W10	May 25 – Jun 7	Integration, benchmark eval, defect fixes	M5: Demo-ready
Pilot Deployment & Final Demo	W11	Jun 8 – Jun 14	Pilot testing, final presentation, closeout	M6: Project closed

6.2 Gantt Chart



7. Budget Summary

Phase	CNY	Notes
Requirement Analysis	300	Reference materials and early validation
System Design	400	Tool setup and environment preparation
Development	1,800	API usage and iteration expenses
Testing	600	Benchmark runs and performance checks
Pilot & Demo	400	Demo environment and presentation materials
Total	**3,500**	Student labor not included

8. Resources

Category	Item	Notes
Human (core)	Xiangyu Xiao 140h · Baoyi Hu 130h · Sirui Da 120h · Qi Lin 118h	Full project participation
Human (support)	Bowen Du (advisor) · Legal Reviewer TBD · Pilot Users	Review and validation

Category	Item	Notes
	(≥20)	
Platform	Tencent AI or approved model service	Model inference and experiments
Software	RAG tooling · backend framework · frontend stack · version control · testing tools	Development and delivery
Facility	Shared dev environment · test/demo deployment environment	Collaboration and verification
Data	Anonymized legal templates · benchmark scenarios · evaluation checklist	Testing and demo

Key provision dates: AI platform + test environment → Mar 29; legal templates + security checklist → Apr 5.

9. Assumptions, Constraints, and Risks

9.1 Assumptions and Constraints

Type	Item	Impact
Assumption	All four members available for full 11-week window	Schedule stability
Assumption	Model-platform access ready by end of design phase	Development timeline
Assumption	Test data prepared without real user data	Testing and demo safety
Assumption	Pilot users available in late May or early June	Validation feasibility
Constraint	Four-student team · fixed window Mar 30 – Jun 14	Scope discipline required
Constraint	Course-project scope · anonymized data only	Prototype-first strategy
Dependency	Model quota · legal templates · test infrastructure · pilot participants	All required for delivery

9.2 Key Risks

Risk	Prob.	Impact	Mitigation	Owner
Manpower shortage / overlapping coursework	High	High	Lock MVP scope; weekly workload review	Xiangyu Xiao
AI hallucination in legal answers	Med	High	RAG grounding, prompt constraints, benchmark audit	Baoyi Hu
Data handling mistakes in testing	Low	High	Anonymized data only; test-data checklist	Xiangyu Xiao
Legal interpretation gaps in outputs	Med	Med	Limit scenarios; legal review for templates	Legal Reviewer
API quota shortage / platform instability	Med	Med	Monitor usage; cache results; fallback demo cases	Qi Lin

9.3 Open Issues

Issue	Status
Legal reviewer naming	Pending confirmation
Final pilot scenario set	To be frozen after requirement analysis
Ownership after course completion	To be clarified at closeout

10. Project Administration

10.1 Communications Plan

Audience	Frequency	Channel
Core team	Weekly	Team meeting and task tracker
Faculty advisor	At milestones	Review meeting and summary update
Legal reviewer	At template checkpoints	Review session and issue list
Pilot users	During pilot testing	Demo, observation, feedback form

10.2 Management Controls

Area	Approach
Scope	Freeze MVP at end of requirement analysis; changes require impact assessment and PM approval
Quality	Benchmarks and checklists defined pre-development; weekly integration checks; no blocker defects before demo
Change	Request → impact assessment → weekly review → approve/defer/reject → update plan

10.3 Implementation and Closure Plan

Stage	Timeline	Output
Integration and testing	By Jun 7	Stable prototype and test results
Pilot and demo preparation	Jun 8–14	Pilot-ready environment and presentation materials
Final closeout	Jun 14	Documentation, archived materials, lessons learned

11. Acceptance Criteria

Item	Criteria
Scope completion	All in-scope modules implemented and demonstrated
Quality threshold	Key MOV targets substantially met; gaps documented
Demo readiness	No blocker defects before final demo
Delivery materials	Pilot checklist and presentation materials completed
Documentation	Final documentation submitted and reviewed

12. References

1. PRC Civil Code (relevant contract and civil provisions)
2. PRC Cybersecurity Law and related data-protection requirements
3. Course requirement materials and team planning notes
4. Approved legal templates and internal review checklist
5. Project README and architecture documentation